

Practice Quiz: Cognition — Shared Understanding Without Words

Part A: Multiple Choice

1. **Distributed knowledge** in a crochet circle means: A) Each member knows exactly the same things B) The group collectively knows more than any individual, and this knowledge combines synergistically C) Knowledge is distributed evenly so no one has more than anyone else D) Only the most experienced member's knowledge matters
2. When a circle member asks "Has anyone worked with this yarn before?", they are: A) Admitting ignorance and should be embarrassed B) Performing a low-cost cognitive action that searches the group's distributed generative models C) Wasting the group's time with a trivial question D) Asking a question that only one person can answer
3. A **shared mental model** about difficulty levels in a circle: A) Is written down in a formal document B) Develops over time through collective experience and allows coordinated communication about project complexity C) Means every member rates difficulty identically D) Is imposed by the most experienced member
4. Nonverbal cognition in a crochet circle is especially powerful because: A) Members prefer not to talk B) Crochet is inherently spatial and tactile, so demonstration communicates what words often cannot C) Language is not used in crochet D) Nonverbal communication is always superior to verbal
5. Group problem-solving in a circle parallels Active Inference because: A) The group ignores evidence and goes with the first suggestion B) Prior beliefs (hypotheses) are confronted with evidence, prediction errors are resolved, and a posterior belief (consensus) emerges C) Only one person thinks while others watch D) The group avoids making any predictions to prevent errors
6. When a circle's shared belief about a technique turns out to be wrong, this represents: A) A fatal flaw in collective cognition B) A collective prediction error that, when

recognized, updates the shared mental model C) Evidence that groups should not think together D) A problem that can never be corrected

7. The cognitive advantage of a crochet circle compared to solitary practice is: A) Nonexistent — individuals think better alone B) The ability to pool partial knowledge from multiple generative models to solve problems no individual could solve alone C) That the group forces everyone to think the same way D) Only relevant for beginners who need help

Part B: Short Answer

1. Describe a scenario where three circle members each contribute partial knowledge to solve a crochet problem that none of them could solve individually. Explain how this parallels distributed Bayesian inference.
2. Give an example of **nonverbal cognition** in a crochet circle — a complex understanding communicated through demonstration without words. Why is this particularly effective for craft knowledge?
3. How does a circle's shared mental model about "what makes good work" develop over time? What role do individual aesthetic preferences play in shaping the collective model?